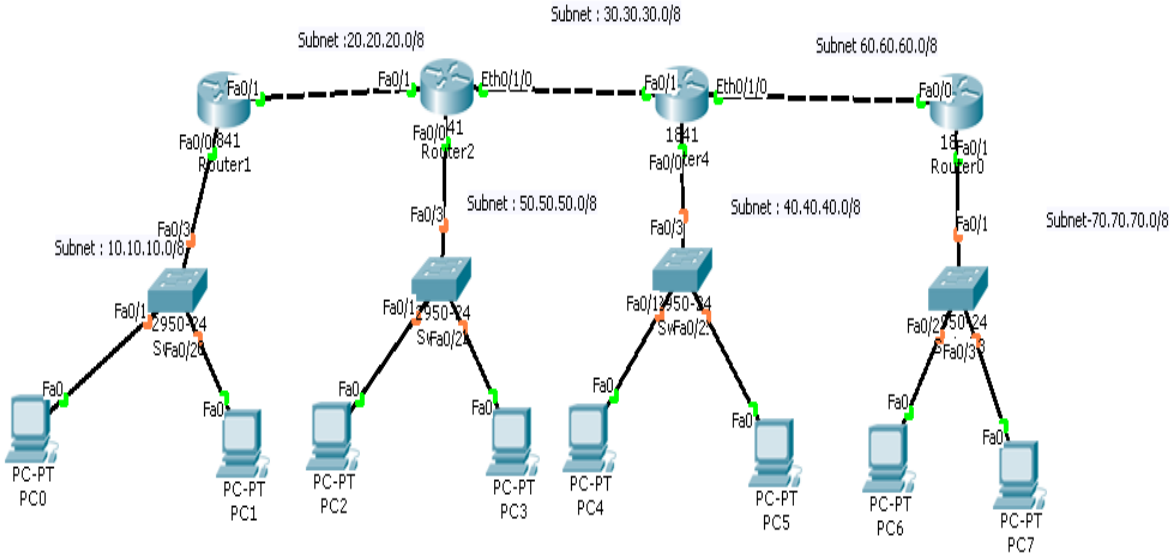
# Dynamic Routing and Analyzing IP Header

## Learning objectives:

- Appreciate Dynamic Routing Protocol ( RIP)
- Understanding the IP header details




### 1. Dynamic Routing Protocol ( RIP)

| | Action | Observation | Reason |
|---|---|---|---|
| | **1.1. In real time mode** | | |
| 1 | Use the topology as shown above. Configure the IP addresses | | |
| 2 | Configure the routing tables of all the routers.<br>**Note , this time you need to select RIP mode** | | |
| | • **Enter the subnet ID of only the subnets attached to each interface of the router**<br>• **Enter only directly connected networks** | | |

| | | | |
|---|---|---|---|
| | • **No need to enter IP address of the next hop** | | |
| 4 | Test the network | **Get the successful communication** | |
| 5 | Use inspection tool ( lens like button in the right ) to click on a router and select Routing table | **Routing table of every router** | **How many subnets entries are created in each router?** |
| | | | |
| | | | |
| | **Action** | **Observation** | **Reason** |
| **1.2. In Simulation mode** | | | |
| 1 | Click power cycle devices | | |
| 2 | Click auto-capture and play | | |
| 3 | Use inspection tool ( lens like button in the right ) to click on a router and select Routing table Give address of router one by one and activate RIP | Keep the table active and wait | |
| | | Initially routing table is blank. | |
| | | Slowly routing table is getting updated. | |

**Open RIP packet and note down the contents**

# *Part B: Analysing IP header*

## IP header analysis with wireshark capture.

| | | |
|---|---|---|
| 7 | **Select Frame 7 again, focus only on IP PDU.** | **Refer trace file 1** |
| 7.1. | What is the source IP address? | **Ans :** |
| 7.2. | Why is the total length shown as 48 bytes? | **Ans :** |
| 7.3 | Can you confirm from that this carries the data given by TCP protocol ? | **Ans :** |
| 7.4 | What is the value of ''More' flag bit ? Justify the value ? | **Ans :** |
| 7.5 | This datagram goes through a series of routers. Assume that there are no mistakes in the configuration of network and there are no errors anywhere in the network. Still this client may get ICMP message from the network. Do you agree? When can this happen? | |

***Open an ICMP request packet and note down the contents***

***Open an ICMP response packet and note down the contents***